Proof. In the spectral representation (via the Spectral Theorem), A is unitarily equivalent to a multiplication operator M_a on some L^2 space such that $A \cong M_a$ and $\chi(A) \cong M_{\chi\circ a}$. In this model, $\sigma(\chi(A))$ is the essential range of $\chi\circ a$ with respect to the underlying measure. But since the measure associated with A is precisely $\mu_x(B) = \langle E_A(B)x, x\rangle$ (for x in an orthonormal basis), the essential range of $\chi\circ a$ equals the set given in the statement (which is defined in terms of E_A) [12, 13]. A direct operator-theoretic proof can be given as follows. If λ lies outside the essential range of χ, then there exists $\delta > 0$ such that $|\chi(\omega)-\lambda| \geq \delta$ for E_A-almost all ω. This implies $(\chi-\lambda)^{-1}$ is essentially bounded on $\sigma(A)$ and thus defines a bounded Borel function. By functional calculus, $(\chi(A) - \lambda I)$ then has a bounded two-sided inverse $(\chi - \lambda)^{-1}(A)$. Hence $\lambda \notin \sigma(\chi(A))$. Conversely, if λ lies in the essential range of χ, we can construct approximate eigenvectors for $\chi(A)$ with eigenvalue λ by concentrating on parts of the spectrum where $\chi(\omega)$ is arbitrarily close to λ. More precisely, for each n choose a Borel set $B_n \subseteq \{\omega : |\chi(\omega)-\lambda| < 1/n\}$ with $E_A(B_n) \neq 0$; pick a unit vector x_n with $\|E_A(B_n)x_n\| \geq \frac{1}{2}\|x_n\|$ (possible since the range of $E_A(B_n)$ is nonzero). Then $y_n := E_A(B_n)x_n/\|E_A(B_n)x_n\|$ is a unit vector such that $\|(\chi(A) - \lambda I)y_n\|^2 = \int_{B_n} |\chi(\omega) - \lambda|^2 \, d\mu_{y_n}(\omega) < 1/n^2$. Thus $\chi(A) - \lambda I$ has approximate kernel vectors, implying $\lambda \in \sigma(\chi(A))$. □

In the context of Theorem 2.2, one important situation is when the limit function χ is exactly the *characteristic function* of the joint peripheral fixed-point set: $\Sigma_{\text{fix}} = \{\lambda \in \sigma(A) : |\lambda| = 1, p_k(\lambda) = \lambda \ \forall k\}$. In that case $\chi(A) = E_A(\Sigma_{\text{fix}})$ is precisely the spectral projection of A onto Σ_{fix}. The next corollary shows that, under mild additional hypotheses on the dynamics of the function iteration, this is indeed the outcome of our process. Essentially, we require that $f^m(\lambda) \to 1$ on Σ_{fix} and $f^m(\lambda) \to 0$ off Σ_{fix}, which means the iteration asymptotically separates the peripheral part of the spectrum from the rest. We will later justify these bullet conditions as a consequence of the peripheral fixed-point property using complex function theory (Lemma 2.6).

Corollary 2.4 (Projection onto a prescribed spectral set)**.** *Suppose, in addition to the standing assumptions, that $p_k(1) = 1$ for each k, and that the function iterates $f^m = f^{\circ m}$ satisfy:*

- $f^m(\lambda) \to 1$ *for all* $\lambda \in \Sigma_{fix}$,
- $f^m(\lambda) \to 0$ *for all* $\lambda \in \sigma(A) \setminus \Sigma_{fix}$,
- $\sup_m \|f^m\|_\infty \leq 1$.

Then, by Theorem 2.1, $A^{(\omega)} = \lim_{m\to\infty} f^m(A) = E_A(\Sigma_{fix})$, which is an orthogonal projection commuting with A.

Proof. Under the bullet assumptions, the pointwise limit function χ of f^m is exactly the characteristic function $\chi = 1_{\Sigma_{\text{fix}}}$ (taking value 1 on Σ_{fix} and 0 elsewhere). Theorem 2.1 then implies strong convergence $f^m(A) \to \chi(A)$, and moreover $\chi(A) = E_A(\Sigma_{\text{fix}})$ is a projection commuting with A. □

Remark 2.5. The above bullet conditions isolate the additional "dynamical" input on the function iterates that is needed to obtain the spectral projection $E_A(\Sigma_{\text{fix}})$. In many concrete choices of the transforms p_k, one can verify these conditions by an elementary analysis of the iterated function $f = p_K \circ \cdots \circ p_1$ acting on the scalar spectral values. For instance, see Lemma 2.6 below and the examples in Section 4.

We emphasize that we have not yet *proven* the bullet conditions of Corollary 2.4 in full generality – indeed, verifying that $f^m(\lambda) \to 0$ for $\lambda \notin \Sigma_{\text{fix}}$ can be a nontrivial problem in complex dynamics. However, a general convergence result is available from the Schwarz–Pick theory of holomorphic self-maps of the disk. Roughly speaking, if the maps p_k are not all the identity, then the composite

f must have an attracting fixed point inside the unit disk (the Denjoy–Wolff point). This yields the following lemma, which provides a rigorous justification of convergence to $\{0, 1\}$ under our assumptions.

Lemma 2.6 (Convergence of f^m to $\{0, 1\}$ under peripheral fixed-point assumptions)**.** *In the normal-case setting, assume in addition that each p_k satisfies the peripheral fixed-point property (so in particular $p_k(1) = 1$). Let $f = p_K \circ \cdots \circ p_1$ as above. Then for each $\lambda \in \sigma(A)$, the sequence $f^m(\lambda)$ converges to a limit $\chi(\lambda) \in \{0, 1\}$. Moreover, $\chi(\lambda) = 1$ if and only if $\lambda \in \Sigma_{fix} = \{\lambda : p_k(\lambda) = \lambda \ \forall k\}$.*

Proof. Consider the holomorphic self-map f of the unit disk $\mathbb{D}$ (note that $\sigma(A) \subseteq \overline{\mathbb{D}}$ by the Schur assumption $\sup |p_k(\lambda)| \leq 1$ and induction). The Denjoy–Wolff Theorem (a foundational result on iterating holomorphic maps on $\mathbb{D}$) guarantees that the sequence $f^m(z)$ converges for every initial $z \in \mathbb{D}$. In fact, one of two cases holds: either (i) f has a fixed point $\tau \in \mathbb{D}$ (with $|\tau| < 1$), in which case $f^m(z) \to \tau$ for all $z \in \mathbb{D}$; or (ii) f has no fixed point in $\mathbb{D}$, in which case the iterates converge to a boundary point (called the Denjoy–Wolff point of f) which must be 1 in our situation. Indeed, in case (ii), classical results (see [21]) state that there is some point ξ on the unit circle $\mathbb{T}$ such that $f^m(z) \to \xi$ as $m \to \infty$ for all $z \in \mathbb{D}$ and $f(\xi) = \xi$ (with an *angular derivative* $\angle f'(\xi) \leq 1$). Since all p_k fix 1 and have the peripheral fixed-point property, it follows that $f(1) = 1$. By the Schwarz lemma for boundary fixed points (the Julia–Wolff–Carathéodory theorem), if f has a boundary fixed point ξ, then ξ is unique and $\xi = \lim_{m\to\infty} f^m(z)$ for every z. Given that 1 is a fixed boundary point of f, it must be this unique attractive boundary point. Thus $f^m(z) \to 1$ for all $z \in \mathbb{D}$ in case (ii).

Now we combine these conclusions with the peripheral fixed-point property. We have established that for each $\lambda \in \overline{\mathbb{D}}$, $\lim_{m\to\infty} f^m(\lambda) =: \chi(\lambda)$ exists. If f has an interior Denjoy–Wolff point $\tau \in \mathbb{D}$ (case (i)), then $\chi(\lambda) = \tau$ for all λ. In this scenario, note that τ must actually equal 0. Indeed, if $\tau \neq 0$ then τ is a fixed point of f *inside the disk* with $\tau \neq 1$, and applying the peripheral fixed-point property to each p_k we see that if $f(\tau) = \tau$ then in fact $p_1(\tau) = \tau, \ldots, p_K(\tau) = \tau$. Hence τ lies in Σ_{fix}, but τ has $|\tau| < 1$ by assumption, contradicting the fact that Σ_{fix} consists of spectral values with $|\lambda| = 1$. Therefore the interior fixed point must satisfy $\tau = 0$. In summary, in case (i) we have $f^m(\lambda) \to 0$ for all $\lambda \in \mathbb{D}$. In case (ii), as discussed, $f^m(\lambda) \to 1$ for all $\lambda \in \mathbb{D}$. We emphasize that case (ii) would occur if and only if f has no fixed point in $\mathbb{D}$, which by the argument above is equivalent to saying that f has no fixed point in $\mathbb{D}$ other than 0 (because any other candidate interior fixed point $\tau \neq 1$ would violate the peripheral fixed-point property, as just shown).

Finally, we identify $\chi(\lambda)$ on $\sigma(A)$. If $\lambda \in \Sigma_{\text{fix}}$, then $p_k(\lambda) = \lambda$ for each k, so $f(\lambda) = \lambda$ and thus λ is a fixed point of f. This means $f^m(\lambda) = \lambda$ for all m, hence $\chi(\lambda) = \lambda$. But since $|\lambda| = 1$ for λ in Σ_{fix} (the "peripheral" part of the spectrum), the peripheral fixed-point property forces $\lambda = 1$ whenever it is fixed by all p_k. (In fact, $\Sigma_{\text{fix}} \subseteq \{z : |z| = 1\}$ by definition, and if $\lambda \in \Sigma_{\text{fix}}$ then taking any p_k such that $p_k(\lambda) = \lambda$ and $|p_k(\lambda)| = 1$ yields $p_k(\lambda) = \lambda$ on $\mathbb{T}$, but since $p_k(1) = 1$, the only way this holds for two distinct unimodular points $\lambda \neq 1$ would be if p_k were the identity map on the arc connecting them, which by analyticity implies p_k is identity, a degenerate case we exclude by assuming not all p_k are trivial.) Therefore the only possible point in Σ_{fix} is $\lambda = 1$. Hence if $\lambda \in \Sigma_{\text{fix}}$, we get $\lambda = 1$ and $\chi(\lambda) = 1$. Conversely, if $\lambda \notin \Sigma_{\text{fix}}$, then λ is not a fixed point of at least one of the p_k, and in particular λ cannot be a fixed point of f. As argued, in that scenario the iterates $f^m(\lambda)$ cannot converge to 1 unless f had no interior fixed point at all (case (ii) above). But if f has no interior fixed point, it must mean that Σ_{fix} was empty (otherwise any $\lambda \in \Sigma_{\text{fix}}$ would be an interior fixed point as we just reasoned, giving a contradiction), so in this scenario indeed $f^m(\lambda) \to 1$ for all λ. However, this scenario is inconsistent with $\lambda \notin \Sigma_{\text{fix}}$ if $|\lambda| = 1$ and $p_k(\lambda)$ remained unimodular for all k. Thus λ must lie strictly inside the disk where one of p_k pushes it to a strictly smaller

modulus. We deduce that $f^m(\lambda) \to 0$ in this case. In summary, if $\lambda \notin \Sigma_{\text{fix}}$, then $\chi(\lambda) = 0$. This completes the proof. $\square$

Summary of normal-case conclusions. Under the Schur and peripheral fixed-point assumptions, Lemma 2.6 ensures that $f^m(\lambda) \to \chi(\lambda) \in \{0, 1\}$ for every $\lambda \in \sigma(A)$, with $\chi = 1_{\Sigma_{\text{fix}}}$. By Theorem 2.1 and Corollary 2.4, we obtain $A^{(\omega)} = E_A(\Sigma_{\text{fix}})$, an orthogonal projection commuting with A. Furthermore, by Theorem 2.2, $\sigma(A^{(m)}) = f^m(\sigma(A))$ at each finite m, and $\sigma(A^{(\omega)}) \subseteq \{0, 1\}$. In other words, the transfinite (countable) limit coincides with the spectral projection of A onto the joint peripheral fixed set, as claimed in the Introduction.

3 Convergence results in the Banach (Ritt/sectorial) case

We now turn to the general (non-normal) setting of a power-bounded or contractive-semigroup operator on a reflexive Banach space. In this setting one typically does not have a functional calculus beyond holomorphic functions, except under additional conditions such as a Ritt or sectorial resolvent bound (which ensure a bounded H^∞ calculus). We therefore impose that A is either:

- a **Ritt operator** on a reflexive Banach space X (meaning $\sigma(A) \subseteq \overline{\mathbb{D}}$, possibly $1 \in \sigma(A)$, with the resolvent condition $\sup_{|z|>1}(|z| - 1)\|(zI - A)^{-1}\| < \infty$ [10, 11]), or
- a generator of a bounded **sectorial analytic semigroup** (meaning $\sigma(A)$ lies inside a sector $\{re^{i\theta} : |\theta| < \Theta\}$ with $\Theta < \pi/2$, possibly $0 \in \sigma(A)$, and appropriate resolvent bounds ensuring an H^∞ calculus).

In either case we assume A admits a bounded H^∞ functional calculus on the spectral domain [9, 11] (cf. [9, 10]). Let $\varphi_1, \ldots, \varphi_K \in H^\infty$ on the disk/sector containing $\sigma(A)$, with $\sup |\varphi_k| \leq 1$ and $\varphi_k(1) = 1$. Define the composite $\Psi := \varphi_K \circ \cdots \circ \varphi_1$. Then each $\varphi_k(A)$ is a well-defined bounded operator with $\|\varphi_k(A)\| \leq C$ for some constant C, and each $\varphi_k(A)$ commutes with A (since the H^∞ functional calculus is an algebra homomorphism). In particular, $\Psi(A) = \varphi_K(A) \cdots \varphi_1(A) = (\varphi_K \circ \cdots \circ \varphi_1)(A)$ is well-defined and commutes with A. We consider the iterative sequence $A^{(n)} = (\Psi(A))^n$ with $A^{(0)} = A$. Our main convergence result in this setting is as follows.

Theorem 3.1 (C – Convergence for Ritt/sectorial operators)**.** Let A be a power-bounded operator on a reflexive Banach space X, with $\sigma(A)$ contained in the closed unit disk (discrete-time case) or inside a sector $\{re^{i\theta} : |\theta| < \Theta\}$ with $\Theta < \pi/2$ (continuous-time case). Assume A is either (i) a Ritt operator of angle $< \pi/2$, or (ii) $-A$ generates a bounded holomorphic C_0-semigroup, and that in either case A has a bounded H^∞ functional calculus on the disk/sector [9, 11]. Let $\varphi_1, \ldots, \varphi_K$ be bounded holomorphic functions on the spectral domain, with $\sup_{\lambda \in \sigma(A)} |\varphi_k(\lambda)| \leq 1$ and $\varphi_k(1) = 1$. Set $\Psi := \varphi_K \circ \cdots \circ \varphi_1$, and assume moreover that $\Psi(A)$ is power-bounded (which holds in particular if $\sup_{\lambda \in \sigma(A)} |\Psi(\lambda)| \leq 1$, by the functional calculus). Then the following hold:

(a) The *mean ergodic* projection $P = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} (\Psi(A))^n$ exists in $B(X)$ (the limit is taken in strong operator topology, equivalently in the weak operator topology on reflexive X). This P is a bounded projection satisfying $P\,\Psi(A) = \Psi(A)\,P = P$ and $\operatorname{Ran} P = \ker(I - \Psi(A)) = \{x \in X : \Psi(A)x = x\}$. Moreover, P commutes with A.

(b) If, in addition, $\sigma(\Psi(A)) \cap \mathbb{T} \subseteq \{1\}$ and 1 is an isolated point of $\sigma(\Psi(A))$ that is a pole of the resolvent of $\Psi(A)$, then in fact $\lim_{n\to\infty}(\Psi(A))^n = P$ (in SOT), i.e. the powers of $\Psi(A)$ converge strongly to the projection P. In particular, $A^{(n)} = (\Psi \circ \cdots \circ \Psi)(A) = (\Psi(A))^n$ converges strongly to P as $n \to \infty$.